Algorithms Design and Analysis - HW II

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Due to: 4/12/2025

Please submit the assignment in pairs, and make sure to include both of your names and ids on the PDF. 5 bonus points are given for a typed out solution, and my respect to whoever writes in LATEX. Make sure to prove each claim you use, unless it was seen in class. You may contact me for every question you have about the H.W.

Problem 1

Given n matrices $A_1, A_2, \ldots, A_n$ with different sizes, we want to compute the product:

$$\prod_{i=1}^{n} A_i$$

Assume all of the shapes match so we can multiply the matrices, and assume multiplying two matrices of sizes $r \times m$ and $m \times k$ takes rmk time.

1. Give an algorithm that runs in $O(n^3)$ which returns the minimum number of operations needed to compute the product. Notice you are not computing the actual product.

   Hint: Give a little thought to how you fill the table.

   If you need a little intuition regarding why multiplying in a different order yields a different number of operations, consider matrices of sizes 1×2, 2×4, 4×1 and try multiplying them in different orders.

2. Write a procedure in pseudo-code that prints the parenthesis such that the number of operations is optimal. For example, matrices of size 1×2, 2×4, 4×1 will yield:

   $$(A_1(A_2A_3))$$

   The input of the procedure is an array P of dimensions, where matrix A_i has dimensions $P[i-1] \times P[i]$.

Problem 2

Given an array A of n real numbers, suggest an efficient algorithm that returns a subsequence (not necessarily contiguous) with maximum sum that does not contain 3 consecutive elements.

Problem 3

Given a **binary** array $A[1 \dots n]$, we define the **bias** of a subarray $A[i \dots j]$ as the number of zeros minus the number of ones in that subarray.

For example, for the array

$$A = [1, 1, 0, 0, 1, 0, 0, 1],$$

the bias of $A[1 \dots 3] = [1, 1, 0]$ is $1 - 2 = -1$.

1. Give a linear-time algorithm that returns the maximum bias in the array A.
2. Using the previous part, provide a linear-time algorithm that returns the indices i, j such that $A[i \dots j]$ obtains the maximum bias.

Hint: (Highlight and copy to reveal)

Problem 4

Given n jobs, each represented as $(startTime[i], endTime[i], profit[i])$, find the maximum profit that can be earned by scheduling jobs such that no two jobs overlap.

For example:
$startTime = [1, 2, 3, 4]$
$endTime = [3, 4, 5, 6]$
$profit = [50, 10, 40, 70]$
$output = 120$

Problem 5

A **shuffle** of two strings X and Y is formed by interspersing their characters into a new string, keeping the characters of each string in the same order.

For example, the string `BANANAANANAS` is a shuffle of the strings `BANANA` and `ANANAS` in several different ways:

`BANANAANANAS`, `BANANAANANAS`, `BANANAANANAS`.

Similarly, the strings `PRODGYRNAMAMMIINCG` and `DYPRONGARMAMMICING` are both shuffles of `DYNAMIC` and `PROGRAMMING`:

`PRODGYRNAMAMMIINCG`, `DYPRONGARMAMMICING`.

Given three strings $A[1 \ldots m]$, $B[1 \ldots n]$, and $C[1 \ldots m + n]$, describe and analyze an algorithm to determine whether C is a shuffle of A and B.